

BUFFER VOLUMES
Liquid buffer: 7 L min
Vapour buffer: 5 L min

SCOPE HAND-OVER
Diamonds (TPXYYYY) mark the
‘last-meter’ hand-over between
in-scope and out-of-scope assets.

LEGEND - ANSI/ISA-5.1 / ISO 10628

PROCESS LINE CLASSES

- 40 K shield - 40 K / 14 bar
- 4.5 K supply - 4.5 K / 3 bar
- 2 K return - 2 K / 27 mbar
- DI water - Cooling water
- 60 K guard - 60 K / 13 bar
- Infrastructure - Scope division
- Instrument air - 6 bar(g)

(process shown grayscale; signals in colour)

INSTRUMENTS (ISA 5.1)

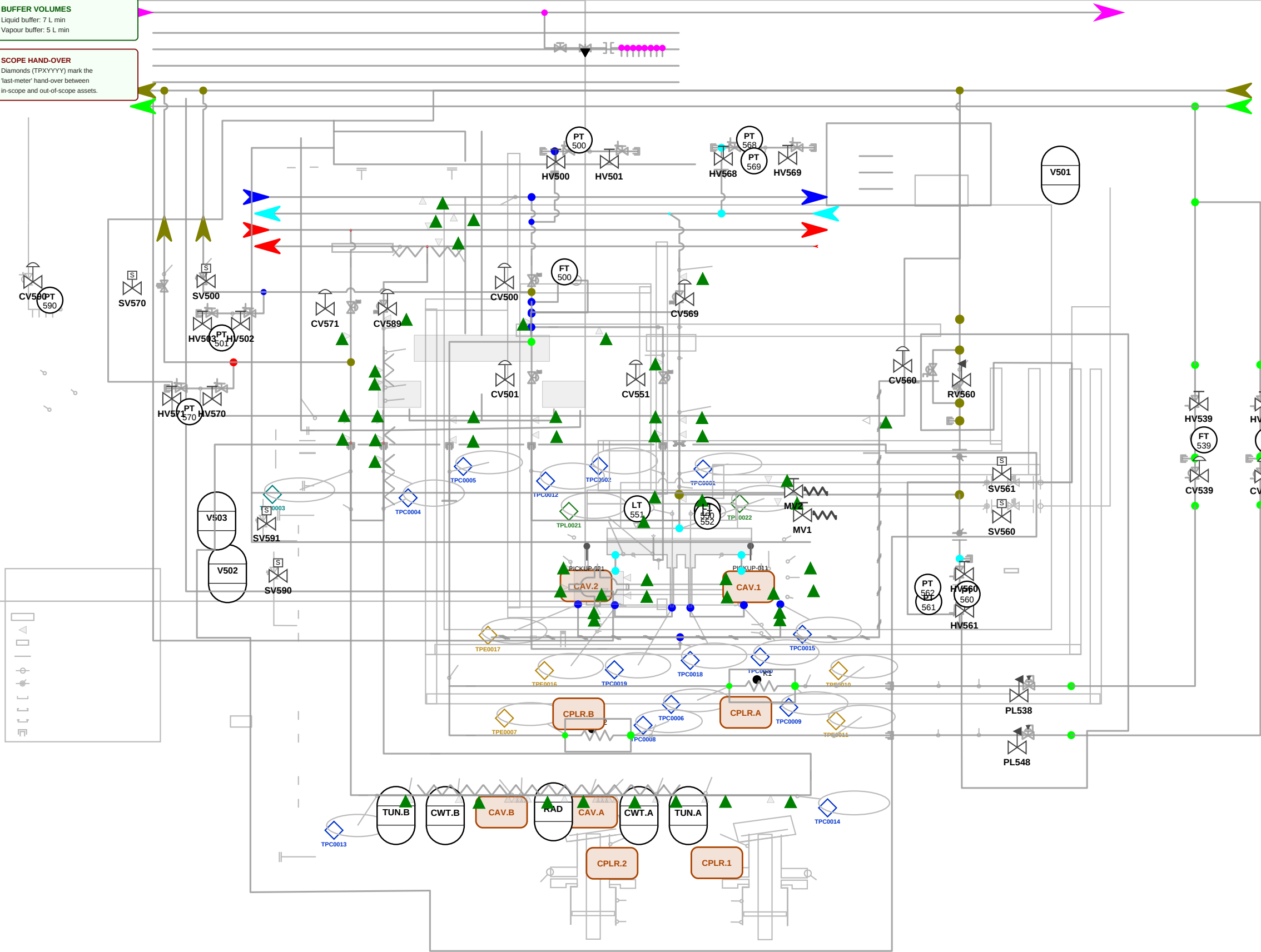
- LB cryo instrument
- RFCELL instrument
- LBI-specific
- Protection / safety (dashed)
- Re-allocated sensor (PZ / CX / PT100)

VALVES & MECHANICAL

- Hand/manual valve
- Solenoid valve
- Control valve
- Relief / limiter
- Bellows (anti thermal short)

SCOPE & CONTROL

- Scope boundary TPXYYYY (last metre)
- Cat: B=Bulk C=Cryogenic E=Electrical H=HVAC L=Liquid S=Steam W=Water
- DIS - Device Interlock System
- Lemo B-series connector (HV/PZ)



PROJECT	MINERVA CryoCell - SCK CEN
TITLE	QCELL / LB - Cryogenic Circuits (40 K / 4.5 K...
SHEET	Sheet1 / 2 - Cryogenic
VERSION	CONTROL-CENTRIC (Version A - signals emphasis...
DRAWING No.	SCK CEN/84836013
STANDARD / SIZE	ISO 10628 / ISA-5.1 - A3 (420x297)